Longitudinal Analysis of Alzheimer's Disease Progression: Modeling Cognitive and Psychiatric Trajectories with Informative Dropout

Anita Kerubo Ogero (2469491) Okolie Tess Eucharia (2468507)
Moses Mburu (2469245) Dorothy Chepkoech (2469284)

Abstract

Alzheimer's disease is characterised by progressive cognitive and neuropsychiatric decline, but longitudinal analysis is complicated by dropout and non-random missingness. We analysed up to six annual follow-up measurements of BPRS (psychiatric symptoms) and CDR-SB (cognitive–functional status) in an Alzheimer cohort, focusing on missingness, disease trajectories, and baseline predictors. Missingness was monotone and strongly related to baseline severity, supporting a missing-at-random (MAR) assumption rather than missing completely at random. For BPRS, a linear mixed model showed a clear increase over time ($\approx 7.2 points/year, p < 0.001$), with higher symptom levels among patients who were older, had poorer ADL, lived in nursing homes, or were not employed. Baseline CDR-SB moderated BPRS progression, with more impaired patients showing slower subsequent worsening. For severe CDR-SB (CDR-SB > 10), logistic mixed models and weighted GEE revealed increasing risk over time and an important role for tau PET. A δ-shift pattern-mixture sensitivity analysis indicated that these conclusions were robust to plausible MNAR departures. Clinically, this suggests our findings give a reliable picture of how psychiatric and cognitive problems progress in Alzheimer's disease, despite substantial loss to follow-up.

Introduction

Alzheimer's disease is the most common cause of dementia and is a progressive neurodegenerative condition that leads to worsening cognition, neuropsychiatric symptoms, and day-to-day functioning (Scheltens *et al.*, 2016). Dementia is a major global health burden, World Health Organization (WHO) estimates that 57 million people were living with dementia in 2021, with nearly 10 million new cases each year, and the burden falls mainly on older adults (World Health Organization, 2025). Because symptoms change gradually and heterogeneously across patients, longitudinal studies are central for describing average progression, quantifying between patient differences, and identifying baseline factors linked to faster or slower deterioration. In Alzheimer research and clinical trials, cognitive functional progression is commonly tracked with the Clinical Dementia Rating–Sum of Boxes (CDR-SB), which captures both cognitive and functional impairment and tends to worsen in a roughly steady way over follow-up in symptomatic disease (Cedarbaum *et al.*, 2013; Williams *et al.*, 2013). Psychiatric symptom burden can be summarised using clinician-rated scales such as the Brief Psychiatric Rating Scale (BPRS), which has been applied in Alzheimer populations to characterise behavioural and psychological symptoms (Ownby *et al.*, 1994).

In this study, we analyse longitudinal data from a cohort of newly diagnosed patients recruited across several clinical centres and followed annually for up to six years. The main outcomes of interest include repeatedly

measured psychiatric symptoms, assessed using the Brief Psychiatric Rating Scale (BPRS), and cognitive–functional status, summarised using the Clinical Dementia Rating Sum of Boxes (CDR-SB). Biomarkers related to Alzheimer's pathology, including amyloid and tau positron emission tomography (PET) measures, were obtained alongside baseline demographic and clinical characteristics, providing a rich multivariate dataset for studying disease progression.

A key challenge in such long-term follow-up studies is the presence of missing data due to dropout, which is particularly common in elderly and clinically vulnerable populations. When missingness is associated with disease severity or functional decline, the observed data may become increasingly selective over time, leading to biased estimates if naive complete-case approaches are used. As emphasised in the longitudinal missing-data literature, valid inference requires statistical models that can account for informative dropout or missingness mechanisms that depend on observed patient history rather than future unobserved outcomes (Molenberghs *et al.*, 2004). This study therefore investigates appropriate longitudinal modelling approaches that remain valid under realistic missingness assumptions and explores the robustness of key clinical conclusions to deviations from the missing at random framework.

The objectives of this analysis are four. First, to describe and explore the extent and patterns of missing data over follow-up. As part of this, we assess whether the data are consistent with missing completely at random (MCAR) by examining associations between missingness and observed patient characteristics and earlier outcomes. Second, to describe and model the evolution of psychiatric symptom severity (BPRS) over time and how it relates to baseline patient characteristics and disease severity, using methods that account for correlation between repeated measurements within the same patient and unbalanced follow-up (Laird and Ware, 1982). Third, to model how the probability of being in the worst cognitive category (CDR-SB > 10) changes over time and how it relates to baseline characteristics and biomarkers, using marginal and subject-specific frameworks for longitudinal binary outcomes (Liang and Zeger, 1986). Fourth, since the missing-at-random (MAR) assumption cannot be verified from the observed data and MAR and missing not at random (MNAR) mechanisms can lead to the same observed patterns, we perform a sensitivity analysis that explores clinically plausible departures from MAR toward missing not at random (MNAR) for the continuous BPRS outcome (Little, 1993).

Methodology

Study design, outcomes, and covariates

We analysed data from a multicentre cohort of patients diagnosed with Alzheimer's disease who were assessed annually for up to six years. Psychiatric symptom severity was measured repeatedly using the Brief Psychiatric Rating Scale (BPRS). Cognitive status was measured using the Clinical Dementia Rating Sum of Boxes (CDR-SB). Two positron emission tomography biomarkers were also assessed annually, amyloid positron emission tomography (amyloid PET) and tau positron emission tomography (tau PET).

For the cognitive endpoint, the continuous CDR-SB score was dichotomised to define a clinically severe category:

$$Z_{ij} = \mathbb{I}\big(\text{CDR-SB}_{ij} > 10\big)\,,$$

where $i = 1, \dots, N$ indexes patients and $j = 0, \dots, J$ indexes the scheduled annual visits, with $j = 0$ baseline.

Baseline characteristics included age, sex, education, body mass index, functional status (activities of daily living), employment status, and living situation. These are denoted collectively by X_i. Baseline disease severity measures are denoted by S_i (including baseline CDR-SB, and baseline biomarker measures where relevant).

Analysis objectives

The analysis had four linked objectives:

1. Describe the amount and structure of missing follow-up data and assess whether missingness depends on observed outcomes or covariates.
2. Characterise the average evolution of BPRS over time and how it relates to baseline characteristics and baseline disease severity.
3. Characterise the evolution over time of the dichotomised cognitive outcome based on CDR-SB.
4. Assess how sensitive the BPRS conclusions are to departures from the primary missing-data assumption.

Missing follow-up data

Notation for missingness

Let Y_{ij} denote BPRS at visit j for patient i.

$$R_{ij} = \begin{cases} 1, & \text{if } Y_{ij} \text{ is observed,} \\ 0, & \text{if } Y_{ij} \text{ is missing.} \end{cases}$$

Let $R_i = (R_{i0}, \dots, R_{iJ})$ denote a patient's observation pattern, and split the full outcome vector as $Y_i = (Y_i^{obs}, Y_i^{mis})$.

Describing missingness patterns

Missingness patterns were first characterised descriptively to determine whether dropout followed a monotone structure or whether intermittent missingness occurred. For each subject, the last observed visit was defined as

$$t_i = \max\{j : R_{ij} = 1\}.$$

A subject was classified as exhibiting monotone dropout if $R_{ij} = 1$ for all $j \leq t_i$ and $R_{ij} = 0$ for all $j > t_i$. Non-monotone (intermittent) missingness was defined as the presence of at least one missing observation followed by a later observed measurement, i.e. the pattern is non-monotone if there exist $j < k$ such that $R_{ij} = 0$ and $R_{ik} = 1$.

These classifications were summarised using (i) the distribution of t_i (retention over time), (ii) the proportion missing at each visit j, and (iii) the frequency of intermittent gaps within R_i.

Assessing dependence of missingness on observed data (showing data are not MCAR)

We examined whether missingness is plausibly independent of observed information (more consistent with MCAR) or related to observed disease severity and covariates (supporting a MAR working assumption). This included descriptive comparisons (for example, baseline BPRS by the number of missed follow-up visits) and a discrete-time model for dropout. The aim is to demonstrate that missingness is not consistent with a completely random observation process and to motivate methods that remain appropriate when dropout depends on observed history (Molenberghs *et al.*, 2004).

A convenient way to formalise dropout is to model the probability of becoming missing at the next visit among those observed at the current visit. Defined as:

$$D_{i,j+1} = 1 - R_{i,j+1},$$

and fit a logistic model of the form

$$\Pr(D_{i,j+1} = 1 \mid \mathcal{H}_{ij}) = \text{logit}^{-1}\big(\alpha_0 + \alpha_1 Y_{ij} + \alpha_2^\top X_i + \alpha_3 j\big)\,,$$

where $\mathcal{H}_{ij}$ denotes the observed history up to visit j.

If dropout is associated with observed outcomes or covariates, this argues against MCAR. Diggle and Kenward called this "random dropout" (dropout depending on observed history), for which likelihood-based inference for the measurement model can proceed without explicitly modelling the dropout process, provided standard ignorability/distinct-parameter conditions hold (Diggle and Kenward, 1994).

Handling missing data in longitudinal models

Complete case analysis

Complete case analysis restricts the analysis to patients with a fully observed BPRS profile across all scheduled visits. In long follow-up studies, this can remove a large proportion of the cohort and shifts the analysis toward patients who remain under observation. When dropout is related to disease course (for example, patients with worse psychiatric symptoms or faster deterioration being more likely to miss later visits), the complete-case estimate of the BPRS trajectory reflects the evolution among stayers rather than the evolution in the full cohort. In that setting, the average yearly change in BPRS can be biased because it is estimated from a selected subgroup, and precision is reduced because many observed BPRS measurements are discarded (Molenberghs *et al.*, 2004).

Last observation carried forward (LOCF) and related single-imputation rules

LOCF replaces each missing BPRS/CDR-SB value after dropout with the last observed BPRS/CDR-SB score for that patient. For a progressive condition, this effectively assumes that a patient's psychiatric symptom level stays constant after the last attended visit. In terms of BPRS evolution, LOCF tends to flatten the estimated mean trajectory at later follow-up because patients who drop out contribute repeated copies of their last score rather than a plausible continued change. This can lead to underestimation (or distortion) of the average change over time, and it treats imputed values as if they were observed when estimating uncertainty, which can make confidence intervals look too narrow (Molenberghs *et al.*, 2004).

Multiple imputation (MI)

Multiple imputation treats the missing outcomes as additional unknown quantities and replaces each missing value with m plausible draws from an imputation model. Each imputed dataset is analysed using the same substantive model, and results are then combined using Rubin's rules so that standard errors reflect both within-imputation variability and the extra uncertainty due to missingness.

Let $\widehat{\theta}^{(k)}$ and $\widehat{\text{Var}}(\widehat{\theta}^{(k)})$ denote the estimate and its variance from imputed dataset $k = 1, \dots, m$. The pooled estimate is

$$\bar{\theta} = \frac{1}{m} \sum_{k=1}^{m} \hat{\theta}^{(k)}.$$

Define the within-imputation variance

$$\bar{U} = \frac{1}{m} \sum_{k=1}^{m} \widehat{\mathrm{Var}}(\hat{\theta}^{(k)}),$$

and the between-imputation variance

$$B = \frac{1}{m-1} \sum_{k=1}^{m} \left(\hat{\theta}^{(k)} - \bar{\theta}\right)^2.$$

The total variance is

$$T = \bar{U} + \left(1 + \frac{1}{m}\right) B.$$

Multiple Imputation can be used for longitudinal outcomes by specifying an imputation model that reflects the repeated-measures structure (time trends, within-patient correlation, and key interactions). In practice, when the proportion missing becomes large (in this study, missingness reaches approximately 60% at later follow-up), MI can become more sensitive to imputation-model specification because a larger fraction of the information about late follow-up trajectories is coming from modelling assumptions rather than directly observed data. Simulation work has shown that the amount of information that can be recovered by MI decreases as missingness increases, and that careful model specification becomes increasingly important at high levels of missingness (Lee and Carlin, 2012).

For these reasons, multiple imputation (MI) was not used as the primary analysis strategy here. Instead, we relied on likelihood-based mixed models for the main longitudinal analyses under a working MAR assumption, and we used a sensitivity analysis to quantify how conclusions change under departures from MAR.

Likelihood-based approaches (primary analyses under a working MAR assumption)

Our primary longitudinal models for BPRS (LMM) and for the dichotomised CDR-SB endpoint (GLMM) are likelihood-based. Under MAR, missingness depends only on observed information and not on unobserved outcomes:

$$p(R_i \mid Y_i, X_i) = p(R_i \mid Y_i^{\mathrm{obs}}, X_i).$$

Under standard ignorability conditions most importantly, that parameters of the measurement model and the missingness model are distinct likelihood inference based on the observed-data likelihood yields valid estimation for the outcome-model parameters without requiring a fully specified joint model for dropout (Rubin, 1976; Molenberghs *et al.*, 2004). MAR cannot be confirmed from the observed data, so we complemented the primary continuous-outcome analysis with a sensitivity analysis that checks clinically plausible departures from MAR.

Longitudinal models

Continuous outcome (BPRS): linear mixed effects model (LMM)

This model describes how psychiatric symptom severity (BPRS) changes over time in the cohort, while allowing different patients to start at different symptom levels and to change at different rates. It accounts for within-patient correlation from repeated measurements and accommodates unequal follow-up due to dropout.

To model the BPRS trajectory, we used the linear mixed effects model specification from Homework 1:

$$\begin{aligned}\text{BPRS}_{it} = {} & (\beta_0 + b_{0i} + u_{0j}) + (\beta_1 + b_{1i})\,\text{time}_{it} \\ & + \beta_2\,\text{sex}_i + \beta_3\,\text{age}_i + \beta_4\,\text{bmi}_i + \beta_5\,\text{job}_i \\ & + \beta_6\,\text{adl}_i + \beta_7\,\text{wzc}_i + \beta_8\,\text{CDR-SB}_{i0} \\ & + \beta_9(\text{time}_{it} \cdot \text{CDR-SB}_{i0}) + \varepsilon_{it}.\end{aligned}$$

fixed effects

- β_0 is the cohort-average baseline BPRS for a reference patient (given the covariate coding), before adding patient- and centre-level deviations.
- β_1 is the cohort-average yearly change in BPRS. A positive value indicates worsening psychiatric symptoms on average over time, while a negative value indicates improvement.
- $\beta_2, \dots, \beta_7$ represent average differences in BPRS associated with baseline characteristics (sex, age, BMI, employment status, activities of daily living, and living situation), holding the other covariates fixed.
- β_8 quantifies the association between baseline cognitive/functional severity (baseline CDR-SB) and baseline psychiatric symptom severity (BPRS).
- β_9 captures effect modification of the time trend by baseline CDR-SB. For example, if $\beta_9 > 0$, patients with higher baseline CDR-SB tend to show a faster increase in BPRS over follow-up.

The model includes patient-specific random effects collected in $b_i = (b_{0i}, b_{1i})^\top$, where b_{0i} represents the patient's deviation from the cohort-average baseline BPRS and b_{1i} represents the patient's deviation from the cohort-average yearly change. The term u_{0j} is a centre/trial-specific random intercept for centre/trial j, capturing systematic baseline differences between centres.

The random effects and residual errors are assumed to follow:

$$\begin{pmatrix} b_{0i} \\ b_{1i} \end{pmatrix} \sim N\left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, D\right), \qquad u_{0j} \sim N(0, \sigma^2_{\text{trial}}), \qquad \varepsilon_{it} \sim N(0, \sigma^2).$$

Here, D quantifies between-patient variability in baseline BPRS and in progression rate and allows correlation between them. The residual term ε_{it} captures within-patient fluctuations and measurement error around the patient-specific trajectory.

This likelihood-based model uses all available BPRS measurements and supports inference under the working MAR/ignorability conditions described earlier (Molenberghs *et al.*, 2004).

Binary outcome (CDR-SB > 10): logistic Generalized Linear Mixed Model (GLMM) (subject-specific)

For the dichotomised cognitive endpoint, the logistic mixed model describes how the probability of being in the severe cognitive/functional category changes over time and with biomarkers, while allowing each patient to have their own baseline risk and their own rate of change. This answers a patient-level clinical question: among patients with the same observed covariates, do individuals still differ in their underlying propensity for severe status, and do they progress at different speeds?

Using the final centred random-intercept and random-slope specified below, the conditional mean model was:

$$\begin{aligned}\operatorname{logit}(\pi_{it}) = \beta_0 &+ \beta_1\,\mathrm{time}_{c,it} + \beta_2\,\mathrm{sex}_i + \beta_3\,\mathrm{age}_{c,i} + \beta_4\,\mathrm{bmi}_{c,i} \\ &+ \beta_5\,\mathrm{adl}_{c,i} + \beta_6\,\mathrm{abpet}_{c,it} + \beta_7\,\mathrm{taupet}_{c,it} \\ &+ \beta_8\Big(\mathrm{time}_{c,it}\cdot\mathrm{taupet}_{c,it}\Big) + b_{0i} + b_{1i}\,\mathrm{time}_{c,it},\end{aligned}$$

where $\pi_{it} = \Pr(Y_{it} = 1 \mid b_i, X_{it})$ and

$$\begin{pmatrix} b_{0i} \\ b_{1i} \end{pmatrix} \sim N\left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \Sigma_b\right).$$

fixed effects (the β's).

Because this is a logistic model, each β is a log-odds effect. Exponentiating gives an odds ratio, interpreted conditionally on the patient's random effects.

- β_0 is the baseline log-odds of being in the severe category at the reference (centred) time for a reference patient, with centred covariates at zero and biomarkers at their reference levels.
- β_1 is the average yearly change in the log-odds of being in the severe category as follow-up time increases (conditional on the random effects).
- β_2 compares males vs females (or the coded reference group) in terms of the log-odds of severe status, holding the other covariates fixed.
- β_3 is the association between age and severe status on the log-odds scale, expressed per one unit increase in centred age (so it represents the effect relative to the sample mean age).
- β_4 is the association between BMI and severe status on the log-odds scale, per one unit increase in centred BMI.
- β_5 is the association between functional status (ADL) and severe status on the log-odds scale, per one unit increase in centred ADL.
- β_6 quantifies the association between amyloid PET (centred, time-varying) and the odds of severe status at a given visit, holding the other covariates fixed.
- β_7 quantifies the association between tau PET (centred, time-varying) and the odds of severe status at a given visit, holding the other covariates fixed.
- β_8 captures effect modification of the time trend by tau PET. If $\beta_8 > 0$, then higher tau PET is associated with a steeper increase over time in the odds of severe status; if $\beta_8 < 0$, the time trend is less steep at higher tau PET.

random effects (the b's).

- b_{0i} is patient i's deviation in baseline log-odds (their underlying propensity to be in the severe category at the reference time, given covariates).
- b_{1i} is patient i's deviation in the time trend (some patients progress faster or slower than average, beyond what is explained by observed covariates and biomarkers).

As a likelihood-based model, the GLMM uses all available outcome data under the same working MAR/ignorability conditions described earlier (Molenberghs *et al.*, 2004).

Population-averaged model: Generalized Estimating Equations (GEE) and MAR-consistent weighting (WGEE)

Clinical target

The GEE model targets population-average effects: it describes how the average probability of severe cognitive status (e.g., $\Pr(Y_{it} = 1)$) changes over time and with covariates in the study population, rather than modelling patient-specific trajectories. In plain terms, it estimates on average in this cohort, how does risk change with time / biomarkers / baseline characteristics? , not how does this particular patient's latent risk evolve?.

Let Y_{it} be the binary outcome for patient i at annual visit t (e.g., $Y_{it} = 1$ if $\text{CDR-SB}_{it} > 10$), and let the covariate vector match Homework 2:

$$X_{it} = \Big(t,\ \text{sex}_i,\ \text{age}_i,\ \text{bmi}_i,\ \text{adl}_i,\ \text{abpet}_{it},\ \text{taupet}_{it},\ t \cdot \text{abpet}_{it},\ t \cdot \text{taupet}_{it}\Big).$$

The marginal mean model is

$$\mu_{it} = \Pr(Y_{it} = 1 \mid X_{it}), \qquad \text{logit}(\mu_{it}) = X_{it}^{\top}\beta.$$

Standard GEE estimating equations

Write $Y_i = (Y_{i1}, \dots, Y_{iT})^{\top}$ and $\mu_i(\beta) = (\mu_{i1}, \dots, \mu_{iT})^{\top}$.
GEE estimates β by solving

$$U(\beta) = \sum_{i=1}^{N} D_i^{\top} V_i^{-1} (Y_i - \mu_i(\beta)) = 0,$$

where $D_i = \partial\mu_i(\beta)/\partial\beta^{\top}$ and V_i is a working covariance for repeated measures within patient i.

$$\text{Corr}(Y_{ij}, Y_{ik}) = \rho^{|j-k|},$$

so correlation decays as visits get further apart. Robust ("sandwich") standard errors are then used to protect inference on β against misspecification of this working correlation (the correlation choice mainly affects efficiency, not the marginal mean target).

Interpretation

Each β is a population-average log-odds effect. For example, β_1 is the average yearly change in the log-odds of being in the severe category, holding the other covariates fixed; $\exp(\beta_1)$ is the corresponding population-average odds ratio per year.

MAR-consistent weighting (WGEE)

With long Alzheimer follow-up, later-visit data can become a selected subset of patients (often healthier / less severe / more adherent). If dropout depends on observed history, unweighted GEE can drift toward describing the "survivors" rather than the original cohort.

Weighted GEE (WGEE) incorporates inverse-probability weights so that observed visits represent both those who remained and those with similar observed history who tended to drop out. Concretely, WGEE replaces $U(\beta)$ by

$$U_w(\beta) = \sum_{i=1}^{N} D_i^\top V_i^{-1} W_i (Y_i - \mu_i(\beta)) = 0,$$

where W_i is diagonal with entries based on the probability that the visit is observed.

Weight model and construction (observation-level weights)

Define the observation indicator

$$R_{it} = \begin{cases} 1, & \text{if } Y_{it} \text{ is observed,} \\ 0, & \text{if } Y_{it} \text{ is missing.} \end{cases}$$

Under a working MAR assumption for weighting, we model the conditional probability of being observed at visit t given observed history up to $t-1$:

$$\Pr(R_{it} = 1 \mid R_{i,t-1} = 1,\ \mathcal{H}_{i,t-1}) = \text{logit}^{-1}\Big(\psi_0 + \psi_1 Y_{i,t-1} + \psi_2^\top X_i + \psi_3 t + \psi_4 \text{abpet}_{i,t-1} + \psi_5 \text{taupet}_{i,t-1}\Big),$$

with fitted values

$$\hat{p}_{it} = \widehat{\Pr}(R_{it} = 1 \mid R_{i,t-1} = 1,\ \mathcal{H}_{i,t-1}).$$

The cumulative probability of being observed through visit t is

$$\hat{\pi}_{it} = \prod_{s=1}^{t} \hat{p}_{is}, \qquad \hat{\pi}_{i0} = 1,$$

and the inverse-probability weight is

$$w_{it} = \frac{1}{\hat{\pi}_{it}}.$$

To reduce the influence of extreme weights (and improve precision), we use stabilized weights,

$$w_{it}^{\text{stab}} = \frac{\hat{\pi}_{it}^{\text{num}}}{\hat{\pi}_{it}^{\text{den}}},$$

where the numerator uses a simpler model (e.g., baseline covariates + time) and the denominator uses the full history model. In practice we implement observation-level weights (a separate weight per measurement), which is the more precise option and is commonly used in standard software workflows.

Interpretation

The mean model $\text{logit}(\mu_{it}) = X_{it}^\top \beta$ stays the same; weighting changes who counts how much. Visits from patients who were unlikely to remain observed (given their observed history) receive more weight, so the fitted population-average trend better reflects the intended cohort rather than only the healthier remainers.

Sensitivity analysis for departures from MAR (continuous outcome: BPRS)

Motivation

In this cohort, follow-up BPRS measurements are often incomplete because participants may drop out due to worsening health, institutionalisation, or death. A practical consequence is that later BPRS values may be observed mainly among "healthier survivors," so analyses that ignore the unobserved post-dropout course can be optimistic about symptom evolution. Our primary analysis therefore works under MAR as a working assumption, but MNAR (dependence on unobserved outcomes) cannot be ruled out from the observed data alone; a sensitivity analysis asks whether our main BPRS conclusions remain stable under clinically plausible MNAR departures.

Notation: outcome, missingness, and dropout time

Let

$$y_i = (y_{i0}, y_{i1}, \dots, y_{iJ})$$

denote subject i's BPRS measurements over planned visits $j = 0, \dots, J$. Define the missingness indicator

$$r_{ij} = \begin{cases} 1, & \text{if } y_{ij} \text{ is observed,} \\ 0, & \text{if } y_{ij} \text{ is missing.} \end{cases}$$

Write $y_i = (y_i^o, y_i^m)$ for the observed and missing components. For monotone dropout, define the dropout time (last observed visit)

$$D_i = \max\{j : r_{ij} = 1\}.$$

Dropout is clinically informative if subjects with worse psychiatric symptoms tend to drop out earlier, because then the observed late follow-up may over-represent people with slower/wetter symptom progression.

Primary analysis under MAR

Our primary BPRS analysis is likelihood-based Linear Mixed Model (LMM). The MAR assumption can be written as

$$f(r_i \mid X_i, y_i^o, y_i^m) = f(r_i \mid X_i, y_i^o),$$

i.e., conditional on covariates and observed outcome history, missingness does not depend further on unobserved BPRS values. This assumption is not testable from the observed data, so we treat it as a working reference model and explicitly probe departures from it.

Pattern-mixture sensitivity analysis using a δ-shift (MNAR perturbation)

A sensitivity analysis should make the MNAR concern concrete for BPRS by stating (i) the direction (are dropouts likely worse than predicted?), (ii) the magnitude (how much worse would they need to be to change conclusions?), and (iii) the timing (late follow-up where missingness is highest).

We implement a pattern-mixture model for sensitivity analysis via a transparent δ-adjustment:

1. Fit the MAR(assumption) LMM and obtain MAR-based predictions for each scheduled visit:

$$\hat{y}_{ij}^{MAR}.$$

2. Impose an MNAR departure after dropout by shifting the unobserved post-dropout BPRS values upward by a constant δ (interpreted clinically as dropouts would have been δ points worse than MAR predicts):

$$y_{ij}^{m}(\delta) = \hat{y}_{ij}^{MAR} + \delta, \qquad j > D_i.$$

3. Sweep δ over a clinically plausible grid (e.g., $\delta \in \{0, 0.5, 1, 2, 3\}$) and refit the same BPRS model on each δ-adjusted completed dataset, tracking the key estimands (notably the time effect and the time-by-baseline-severity interaction).

Interpretation

- $\delta = 0$ reproduces the MAR-based analysis.
- $\delta > 0$ encodes the clinically plausible scenario that participants who discontinue follow-up would, on average, have had higher (worse) BPRS than predicted from their observed history.
- If the estimated BPRS time trend and its modification by baseline severity remain essentially unchanged over reasonable δ values, we report the conclusions as robust to MNAR departures.
- If conclusions change materially for small-to-moderate δ values, we report that inference about BPRS evolution is sensitive to assumptions about unobserved post-dropout outcomes, and we describe the approximate δ threshold at which conclusions change.

Results

Exploratory Analysis

The exploratory analysis showed that baseline BPRS values increased systematically with the number of missed follow-up visits, and the visit-specific proportion of missing BPRS measurements increased monotonically from 0% at baseline to around 60% by year 6 (Figure 4). Additionally, BPRS data showed a linear increase in mean symptom severity over time, as illustrated by the LOESS-smoothed trend (Figure 1), with relatively stable dispersion of individual scores at each visit. Examination of squared residuals and the empirical semi-variogram (Figure 2) indicated that the residual variance was broadly constant across follow-up, while the semivariogram rose with time lag, consistent with strong positive within-subject correlation that decays as measurements are further apart in time features. This support the use of random intercepts and slopes in a linear mixed-effects framework.

In addition to the BPRS patterns, the exploratory summaries of the binary CDR-SB outcome provide important insight into cognitive functional deterioration over time. As shown in (Figure 3) (left panel), the distribution of CDR-SB categories shifts markedly over the follow-up period, with the proportion of patients classified as severe (CDRSB > 10) increasing steadily from approximately 25–30% at earlier visits to more than 60% by

year six. This monotonic rise indicates progression of cognitive impairment in the cohort. The sex-stratified trajectories (Figure 3) (right panel) showsimilar patterns for males and females, with only differences in early visits and near-convergence at later follow-up, suggesting that sex does not modify the probability of crossing into severe CDR-SB status. Overall, these descriptive patterns are consistent with expected Alzheimer's disease progression and support the use of longitudinal binary models to capture both the increasing probability of severe impairment and the potential influence of baseline covariates and biomarkers. Moreover, the clear monotone nature of the transition from non-severe to severe status reinforces the suitability of models that account for repeated measures and informative dropout.

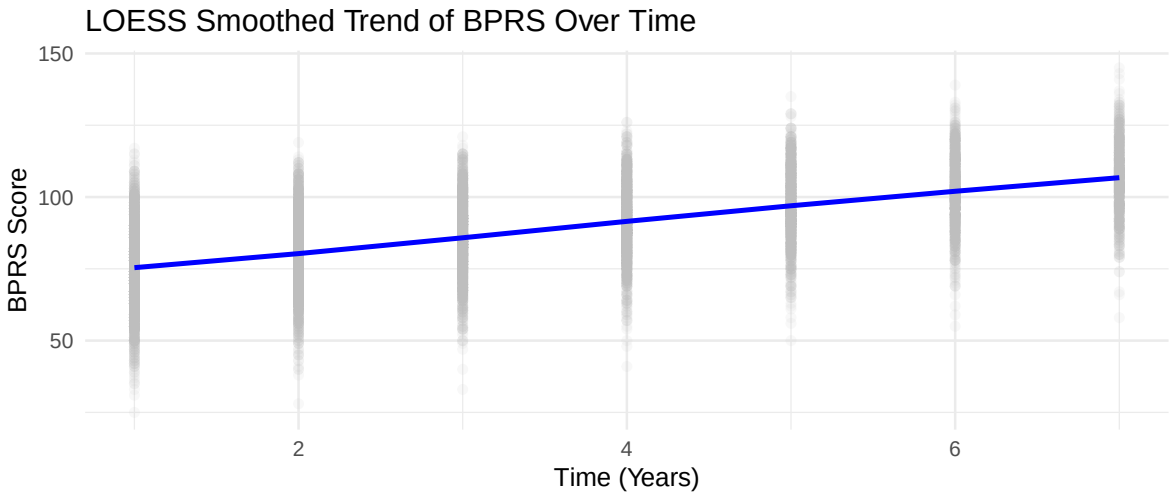

Figure 1: LOESS smoothed trend of BPRS over time.

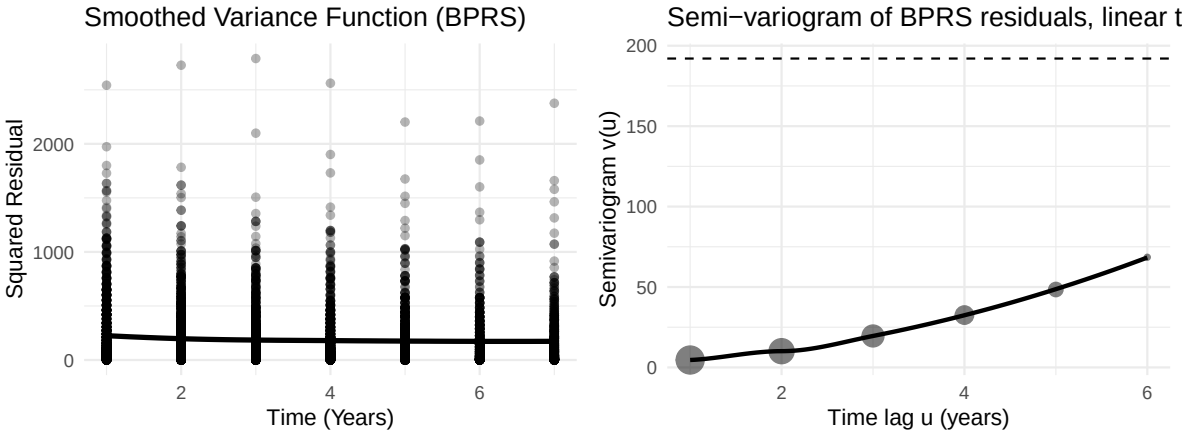

Figure 2: Smoothed Variance Function (left) and Semi-variogram (right) of BPRS Residuals.

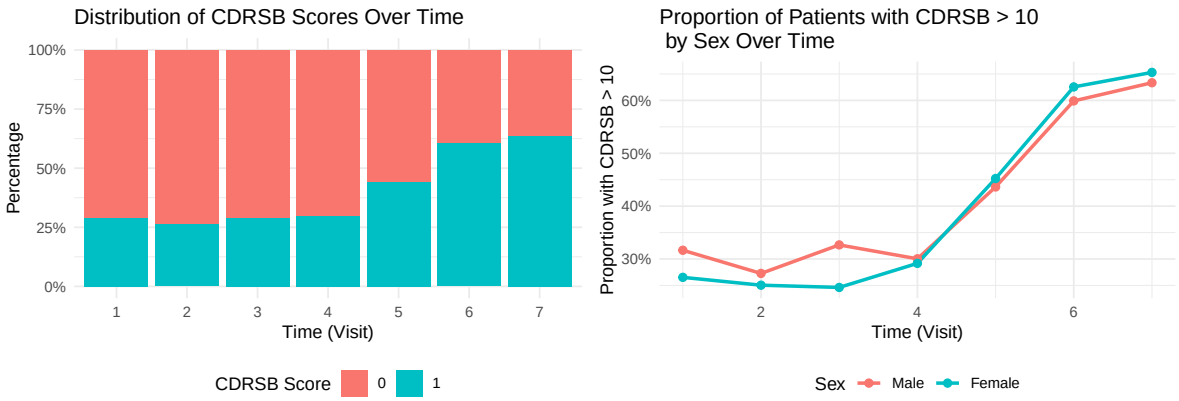

Figure 3: Distribution of CDRSB Scores Over Time and by sex

Investigation of Missingness Mechanisms

Missingness and Dropout Patterns

Among the $N = 1253$ patients, 511 (40.8%) had complete BPRS follow-up across all seven scheduled visits (baseline and six annual follow-ups), whereas 742 (59.2%) exhibited monotone dropout, with no evidence of intermittent missingness (Table 1). At baseline, BPRS was observed for all patients (0% missing), but the proportion of missing measurements increased steadily over time: 11.7% at year 1, 19.1% at year 2, 27.6% at year 3, 38.0% at year 4, 48.0% at year 5, and 59.2% at year 6 (Table 2). Thus, later visits are based on a progressively smaller and more selected subset of the original cohort.

The most frequent missingness patterns were all monotone. In addition to the complete pattern (observed at all visits), common dropout patterns included early dropout after baseline (observed only at baseline, then missing from year 1 onward) and dropout at intermediate time points (e.g., observed through year 4 and missing thereafter), confirming that once patients become missing they do not re-enter follow-up (Table 3).

Table 1: Missingness Patterns in BPRS Measurements Over Time

Missingness pattern	Number of patients	Percent
Complete follow-up	511	40.78
Monotone dropout	742	59.22

Baseline BPRS was positively associated with the extent of subsequent missingness, patients with higher baseline BPRS scores tended to miss a larger number of follow-up measurements (Figure 4). In parallel, the percentage of missing BPRS measurements increased approximately monotonically with time (right panel of Figure 1). Together, these findings indicate that missingness is driven by a monotone dropout process that is time-dependent and related to observed baseline severity, inconsistent with a missing-completely-at-random mechanism and supporting a missing-at-random assumption as a working basis for the longitudinal modelling.

Table 2: BPRS observed and missing measurements by scheduled visit

Visit	Observed	Missing	Percent missing
Baseline	1253	0	0.0
Year 1	1106	147	11.7
Year 2	1014	239	19.1
Year 3	907	346	27.6
Year 4	777	476	38.0
Year 5	652	601	48.0
Year 6	511	742	59.2

Table 3: Most Frequent Missingness Patterns in BPRS Measurements Over Time by Missingness Group

Group	Baseline	Year 1	Year 2	Year 3	Year 4	Year 5	Year 6	Number of patients	Percent
Complete follow-up	O	O	O	O	O	O	O	511	100.00
Monotone dropout	O	M	M	M	M	M	M	147	19.81
Monotone dropout	O	O	O	O	O	O	M	141	19.00
Monotone dropout	O	O	O	O	M	M	M	130	17.52

Baseline BPRS by Number of Missed Visits and Percentage Missing Over Time

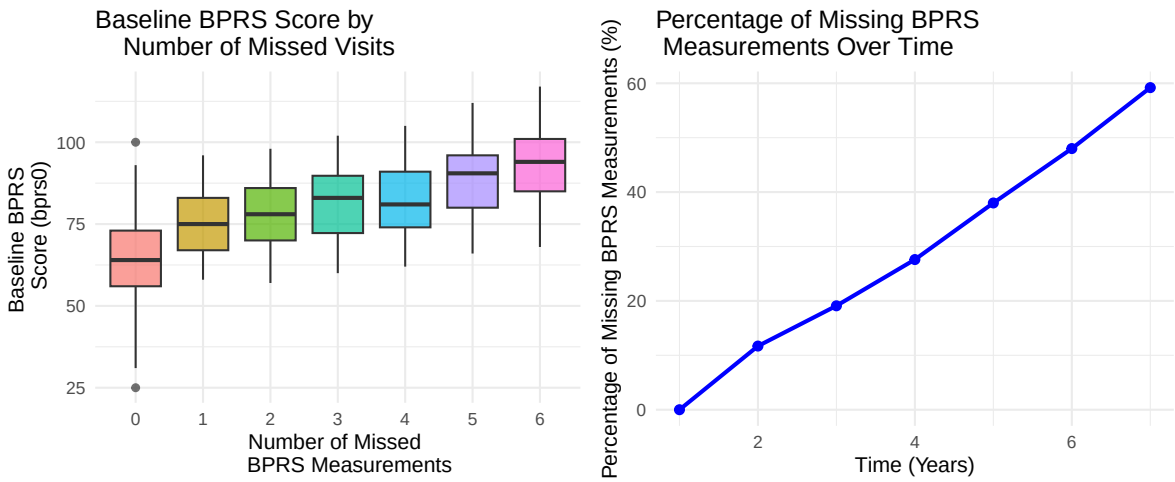

Figure 4: Missingness Analysis: Baseline BPRS by Number of Missed Visits (left) and Percentage Missing Over Time (right)

Summary of MAR-valid model choices

Table 8 summarises the three models implemented in this analysis, together with their outcome scale, inferential target, and the justification for MAR validity.

Table 8: Summary of MAR-valid model choices

Outcome	Scale	Model	MAR Validity Mechanism	Target Estimand
BPRS	Continuous	Linear mixed model (LMM)	Likelihood-based inference remains valid under MAR (ignorable missingness)	Subject-specific mean trajectory
CDR-SB > 10	Binary	Logistic GLMM	Likelihood-based, random-effects model valid under MAR	Subject-specific odds of severe status
CDR-SB > 10	Binary	Weighted GEE	Inverse probability weighting ensures consistency under MAR	Population-averaged risk trajectory

Continuous outcome (BPRS): Linear mixed-effects model under MAR

A linear mixed-effects model (LMM) with patient-specific random intercepts and slopes and a trial-level random intercept was fitted to all available BPRS measurements. Under the MAR/ignorability assumptions, this likelihood-based model provides valid inference using the observed-data likelihood.

The fitted model showed a strong and highly significant increase in psychiatric symptom severity over time. The estimated yearly increase in BPRS was 7.220 points per year (95% CI: 7.141–7.299, $p < 0.001$), confirming a monotonic worsening over follow-up. Several baseline covariates contributed meaningfully to the level of BPRS: age ($p < 0.001$), BMI ($p < 0.001$), employment status ($p < 0.001$), and nursing home residence ($p < 0.001$). Female sex showed a borderline effect ($p = 0.067$). ADL and baseline CDR-SB were not significant main effects.

A clinically important moderation effect was observed for baseline cognitive severity. The interaction between baseline CDR-SB and time was significantly negative (estimate $= -0.022$, $p < 0.001$), indicating that patients with worse baseline CDR-SB exhibited slower increases in BPRS over time. This suggests a convergence or ceiling effect in longitudinal psychiatric symptom trajectories.

The random-effects structure reflected substantial between-patient heterogeneity. The SD of the random intercept was 0.898, and the SD of the random slope was 0.828, with a strong negative correlation (-0.612), indicating that patients starting with higher BPRS tended to progress more slowly. The residual SD was 1.643, consistent with the variance patterns observed in exploratory analysis.

These results demonstrate that the selected LMM captures the major longitudinal trends, covariate effects, and within-subject correlation structure in BPRS, while remaining valid under MAR.

Table 5: Linear Mixed Model Results with with patient Specific Random Intercept and Slope and trial Random Intercept

Parameter	Estimate [95% CI]	SE	Statistic	p-value
Intercept	-60.293 [-68.475, -52.111]	4.169	-14.461	<0.001
Time (years)	7.220 [7.141, 7.299]	0.040	179.316	<0.001
SexFemale	-0.161 [-0.333, 0.011]	0.088	-1.837	0.067
Age (years)	1.697 [1.619, 1.776]	0.040	42.364	<0.001
BMI (kg/m^2)	0.178 [0.125, 0.232]	0.027	6.537	<0.001
EmployedPaid Job	-5.136 [-6.713, -3.558]	0.804	-6.386	<0.001
ADL	0.065 [-0.166, 0.296]	0.118	0.555	0.579

ResidenceNursing Home	1.915 [1.723, 2.108]	0.098	19.565	<0.001
Baseline CDR-SB	0.002 [-0.013, 0.017]	0.008	0.263	0.792
Time (years):Baseline CDR-SB	-0.022 [-0.030, -0.014]	0.004	-5.248	<0.001
SD Intercept	0.898			
Cor Intercept x Time (years)	-0.612			
SD Time (years)	0.828			
SD Intercept	2.829			
SD Observation	1.643			

Evolution of CDRSSB > 10 over time : Logistic GLMM (subject-specific model)

A random-effects logistic regression model (GLMM) was used to estimate subject-specific odds of entering the severe cognitive/functional category (CDR-SB > 10). This model includes both random intercepts and random slopes, enabling individual-level deviations in baseline risk and disease progression rate, and is valid under MAR due to its likelihood-based formulation.

The yearly evolution was strong and statistically significant. The odds of severe CDR-SB increased almost two-fold per year (OR = 1.986, 95% CI: 1.783–2.211, $p < 0.001$). Covariate effects were less pronounced than in the continuous BPRS model. Sex, age, BMI, ADL, and amyloid PET were not significant predictors of severe status in the presence of random effects.

Tau PET showed a significant time-modifying effect: the interaction term time × tau PET was associated with a 47% increase in the yearly odds of severe status (OR = 1.473, 95% CI: 1.062–2.044, $p = 0.020$), suggesting that tau burden accelerates progression to severe cognitive impairment.

Random effects revealed notable between-patient heterogeneity (SD of random intercept = 0.516), confirming the appropriateness of the subject-specific modelling framework.

Table 6: Logistic regression mixed effect model results with patient specific random intercept and slope

Parameter	Estimate [95% CI]	SE	Statistic	p-value
Intercept	0.438 [0.382, 0.504]	0.031	-11.663	<0.001
Time (years)	1.986 [1.783, 2.211]	0.109	12.521	<0.001
SexFemale	0.924 [0.778, 1.098]	0.081	-0.895	0.371
Age (years)	0.987 [0.965, 1.010]	0.011	-1.102	0.270
BMI (kg/m²)	1.008 [0.970, 1.048]	0.020	0.412	0.680
ADL	0.999 [0.967, 1.033]	0.017	-0.032	0.974
Amyloid PET	0.857 [0.634, 1.158]	0.132	-1.004	0.315
Tau PET	1.073 [0.584, 1.970]	0.333	0.227	0.821
Time (years):Tau PET	1.473 [1.062, 2.044]	0.246	2.319	0.020
SD Intercept	0.516			
Cor Intercept x Time (years)	-0.533			
SD Time (years)	1.344			

Population averaged GEE Results with MAR-consistent Weighting

To obtain population-averaged effects under MAR, a weighted generalized estimating equations (WGEE) model was fitted using stabilized inverse-probability-of-observation weights(Figure 5) derived from a discrete-time dropout model. This weighting is required because dropout depends on observed history and standard (unweighted) GEE would not yield MAR-consistent estimates.

The marginal time effect was not statistically significant (OR = 0.809, 95% CI: 0.532–1.230, p = 0.321), in contrast to the strong positive time effect seen in the GLMM. This difference reflects the distinct estimands of GEE (population-average) versus GLMM (subject-specific): the latter is expected to produce larger effects due to non-collapsibility of logistic models.

Tau PET again showed a significant positive time-interaction (OR = 1.372, 95% CI: 1.101–1.710, p = 0.005), consistent with findings from the GLMM. Amyloid PET did not show significant main or interaction effects. Other covariates (sex, age, BMI, ADL) showed no statistically meaningful population-average associations with severe status.

The AR(1) working correlation estimate was 0.315, consistent with moderate within-subject dependence across annual visits. Weight diagnostics indicated stable weights without extreme values, supporting the reliability of the weighted GEE estimates.

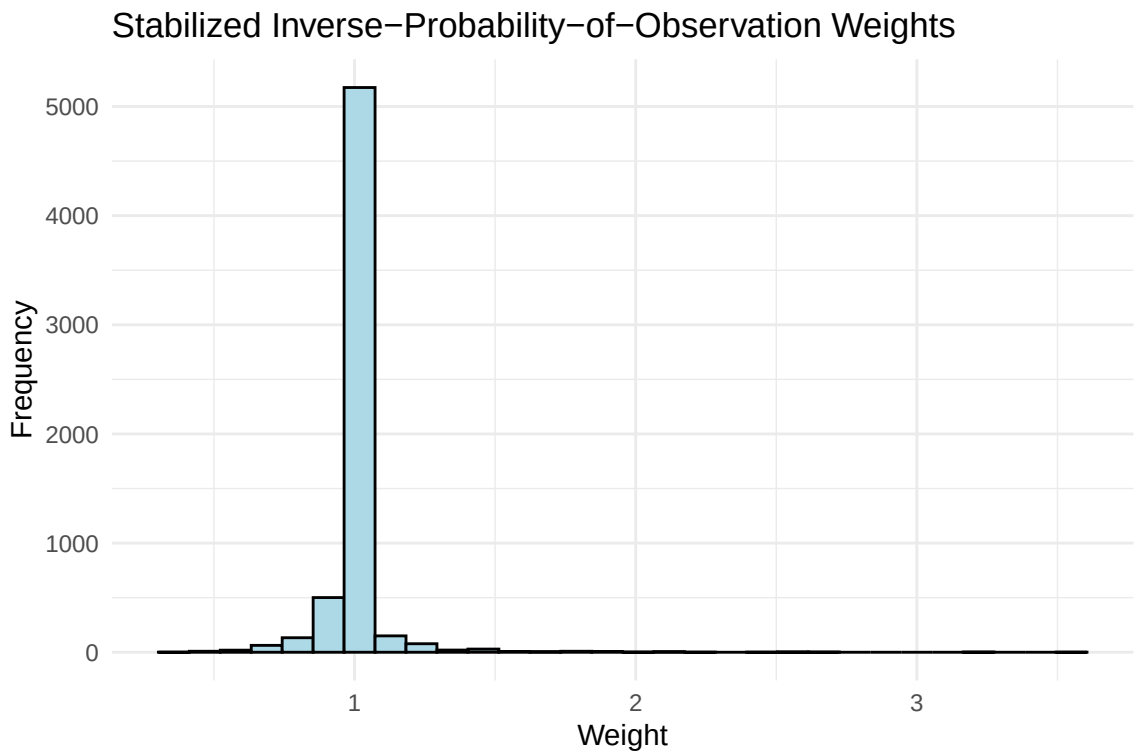

Figure 5: Distribution of stabilized inverse-probability-of-observation weights used in the weighted GEE analysis for CDR-SB > 10

Table 7: Weighted GEE Model Results with MAR-consistent Weighting for CDR-SB > 10

Parameter	Estimate [95% CI]	SE	Statistic	p-value
Intercept	1.759 [0.155, 19.997]	1.240	0.208	0.649
Time (years)	0.809 [0.532, 1.230]	0.214	0.984	0.321
SexFemale	0.911 [0.813, 1.022]	0.059	2.504	0.114
Age (years)	0.990 [0.976, 1.004]	0.007	2.135	0.144
BMI (kg/m²)	1.009 [0.984, 1.034]	0.013	0.485	0.486
ADL	0.987 [0.967, 1.008]	0.011	1.409	0.235
Amyloid PET	1.274 [0.867, 1.873]	0.197	1.518	0.218
Tau PET	0.379 [0.119, 1.208]	0.591	2.693	0.101
Time (years):Amyloid PET	0.938 [0.855, 1.029]	0.047	1.819	0.177
Time (years):Tau PET	1.372 [1.101, 1.710]	0.112	7.907	0.005

Agreement between exploratory analysis and model-based results

Continuous outcome (BPRS): linear mixed-effects model

The exploratory analysis of BPRS showed three main empirical features: (i)a clear, approximately linear increase in the mean BPRS over time; (ii) strong within-subject dependence, as indicated by an increasing semi-variogram with time lag; and (iii) some indication that variability changed with follow-up, with larger dispersion at later visits based on the raw data plots and empirical variance estimates.

The fitted linear mixed-effects model for BPRS is broadly consistent with these empirical findings. The strong positive fixed effect of time confirms the increasing mean trajectory suggested by the LOESS-smoothed curves. The random intercept and random slope components induce within-subject correlation and allow subjects to differ in both baseline level and rate of change, which matches the heterogeneous individual trajectories observed in the spaghetti plots. The negative correlation between random intercept and slope estimates is also compatible with the empirical impression that patients with higher baseline BPRS sometimes show slower subsequent increases.

The empirical variance function from(Figure 2) does not need to coincide exactly with the variance implied by the mixed model. Empirical variances at later visits are calculated from fewer and more selected subjects due to dropout, and are therefore subject to increased sampling variability and potential selection effects. In contrast, the mixed model imposes a parsimonious covariance structure, with a constant residual variance σ^2 and smoothly varying marginal variance driven by the random effects. As a consequence, moderate discrepancies between the empirical variance function and the fitted variance function are expected and do not, on their own, indicate model failure. What matters for the present inferential goals is that the mixed model adequately reproduces the dominant mean trend, the main covariate effects, and the within-subject correlation structure. On this basis, the LMM results are in good qualitative agreement with the exploratory analysis.

Binary outcome (CDR-SB > 10): logistic GLMM (subject-specific model)

For the binary severe-status outcome $Z_{ij} = \mathbb{I}(\text{CDRSB}_{ij} > 10)$, the exploratory analysis in suggested that the prevalence of severe CDR-SB increased with follow-up and tended to be higher among patients with worse baseline severity and less favourable functional or living-status profiles. These raw patterns are reflected in the fitted logistic GLMM: the model yields increasing subject-specific odds of severe status over time and higher

conditional odds among patients with poorer baseline profiles or elevated biomarker levels, in line with the exploratory trends.

Because Z_{ij} is binary, the mean–variance relationship is constrained by the binomial form

$$\mathrm{Var}(Z_{ij} \mid \text{covariates}) = p_{ij}(1 - p_{ij}),$$

where $p_{ij} = \Pr(Z_{ij} = 1 \mid b_{0i})$ is determined by the logistic link and the random effect b_{0i}. The logistic GLMM therefore does not aim to match the empirical variance at each time point separately, but rather to provide a coherent subject-specific model for the conditional probabilities p_{ij} and the induced within-subject correlation. Differences between the empirically observed proportions (and their binomial variances) and those implied by the GLMM are not unexpected, especially at later visits where sample sizes are smaller and dropout is more pronounced. As with the continuous case, the key question is whether the model captures the main temporal trend and covariate associations seen in the exploratory plots; in this sense, the GLMM results are in qualitative agreement with the initial descriptive findings.

Binary outcome (CDR-SB > 10): weighted GEE (marginal model)

At the marginal level, the unadjusted proportions of severe CDR-SB over time suggested an increasing population-average risk of severe status. The weighted GEE (WGEE) model targets these population-averaged effects while correcting for MAR dropout via inverse-probability weights. After adjustment for baseline covariates and weighting for observation probability, the estimated marginal time effect was smaller (and in some specifications not statistically significant), which differs from the strong increasing trend suggested by the crude proportions and by the subject-specific GLMM.

This apparent discrepancy does not necessarily indicate inconsistency between the exploratory analysis and the WGEE model; rather, it reflects (i) the difference between subject-specific and marginal estimands in logistic models, and (ii) the impact of adjusting for covariates and dropout-related selection. The crude proportions do not account for changes in the composition of the risk set over time or for the fact that patients at higher risk of severe status are more likely to drop out. The WGEE, by construction, reweights the observed data to represent what would have been seen under a MAR mechanism in the full cohort, and thus can attenuate or modify the apparent time trend suggested by naïve descriptive summaries.

In summary, for all three models, the main qualitative features from the exploratory analysis progression of psychiatric and cognitive impairment over time, and higher severity among clinically vulnerable subgroups are recovered by the fitted models. Where discrepancies arise (for example, between empirical and model-based variance functions, or between crude proportions and weighted marginal estimates), they can be explained by sampling variability, dropout-induced selection, and the different targets and assumptions of the parametric models, rather than by fundamental disagreement between the data and the modelling framework.

Sensitivity analysis for BPRS under a pattern-mixture δ-shift MNAR model

To assess the robustness of the longitudinal BPRS conclusions to potential missing-not-at-random (MNAR) departures, a pattern-mixture δ-shift sensitivity analysis was performed. For each subject, MAR-based predictions $\hat{y}_{ij}^{\mathrm{MAR}}$ from the primary LMM were obtained at all scheduled visits. For post-dropout occasions ($j > D_i$, where D_i denotes the last observed visit for subject i), missing BPRS values were then replaced by

$$y_{ij}^{\mathrm{mis}}(\delta) = \hat{y}_{ij}^{\mathrm{MAR}} + \delta,$$

with $\delta \in \{0, 0.5, 1, 2, 3\}$ representing increasingly severe MNAR scenarios in which dropouts would have had systematically worse BPRS than predicted under MAR. For each δ, the same LMM was refitted and the key estimands (the time effect and the time-by-baseline-severity interaction) were tracked.

The (Figure 6) shows the estimated time effect and time-by-baseline-severity interaction, with 95% confidence intervals, as functions of δ. The estimated yearly time effect at $\delta = 0$ (the MAR analysis) is strongly positive, and the corresponding confidence interval does not include zero. As δ increases from 0 to 3, the estimated time effect increases only modestly and its confidence intervals remain narrow and entirely above zero, indicating that the conclusion of a pronounced increase in BPRS over follow-up is insensitive to the considered MNAR shift.

Similarly, the time-by-baseline-severity interaction remains negative across all δ values. The estimated interaction lies in a relatively narrow band (approximately between -0.030 and -0.020) and its 95% confidence intervals do not cross zero for any of the examined δ values. This corroborates the main MAR-based finding that higher baseline CDR-SB is associated with a less steep increase in BPRS over time, and shows that this moderating effect is robust to moderate MNAR perturbations of the unobserved post-dropout outcomes.

The random-effects structure of the LMM remains stable across δ values, with a negative correlation between random intercept and random slope (approximately -0.53) and a similar standard deviation of the time slope (around 1.34), indicating that the pattern-mixture perturbations affect primarily the fixed effects rather than the underlying subject-specific variability.Taken together, these results suggest that the main conclusions namely, that BPRS increases over time and that this increase is moderated by baseline cognitive/functional severity are robust to a range of clinically plausible MNAR departures encoded by the δ-shift pattern-mixture model.

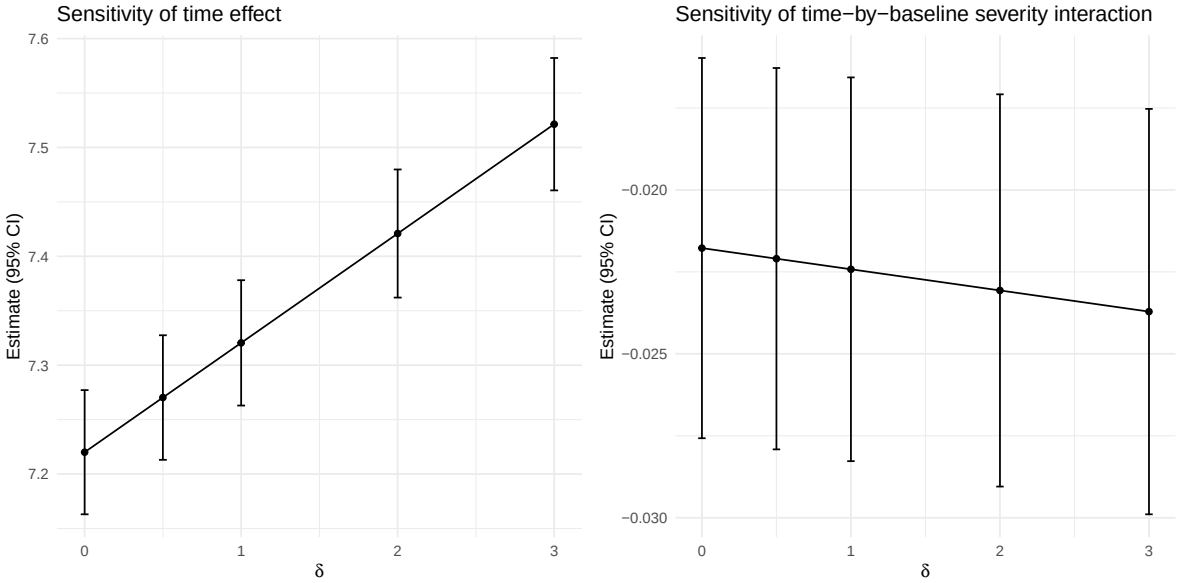

Figure 6: Sensitivity analysis for BPRS time effect and time-by-baseline-severity interaction under a pattern-mixture δ-shift MNAR model

Discussion

In this longitudinal study of patients with Alzheimer's disease, we followed psychiatric symptoms (BPRS) and cognitive–functional status (CDR-SB) over several years and examined how these trajectories were influenced by baseline patient characteristics. Because many patients did not complete all visits, a major focus was to use methods that give clinically meaningful results despite dropout, and to check how sensitive our conclusions are to assumptions about the missing data.

Dropout pattern interpretation

We observed a clear dropout pattern: many patients stopped attending visits as follow-up progressed, and those who dropped out tended to be more impaired at baseline (worse ADL, higher CDR-SB, more often in nursing homes). This pattern is fully consistent with earlier Alzheimer cohorts reporting that attrition is strongly associated with clinical severity and functional status (Schneider, Insel and Weiner, 2011). This means that the patients we see at later visits are, on average, healthier than the original cohort. For this reason, simple analyses that only use "complete cases" (those who finished all visits) would be misleading and would systematically underestimate the true level of symptoms in the original population.This is aligned with methodological guidance for longitudinal dementia studies emphasizing that dropout must be handled using likelihood-based or weighted approaches (Henderson, Diggle and Dobson, 2000; Hogan, Roy and Korkontzelou, 2004).

Instead, we used methods that remain valid if missing visits depend on already observed clinical information (for example, baseline severity and earlier scores), but not on future unobserved outcomes. This is the standard "missing at random" (MAR) assumption and is consistent with what we see in the data: dropout is clearly related to observed severity and functioning.

Longitudinal course of psychiatric symptoms (BPRS)

Clinically, the most important finding for BPRS is that psychiatric symptoms worsen substantially over time. On average, BPRS increased by about 7 points per year, and this result was very precise and highly statistically significant. This trajectory aligns with earlier findings that neuropsychiatric symptoms worsen as Alzheimer's progresses (Lyketsos *et al.*, 2011) and may cluster into predictable patterns (Garre-Olmo *et al.*, 2010).Patients with more functional impairment at baseline (worse ADL), higher BMI, no paid job, and those living in nursing homes started out with higher BPRS levels and tended to remain more symptomatic over time, which matches clinical expectations, a relationship also well-documented in clinical studies linking functional deterioration to neuropsychiatric severity (Tekin *et al.*, 2001).

Interestingly, patients with more severe cognitive/functional impairment at baseline (higher CDR-SB) showed a less steep increase in BPRS over time. This may reflect a ceiling effect or a shift in clinical presentation: patients who are already very impaired cognitively may have less room for further increase in observed psychiatric symptoms, or they may become less able to express or manifest these symptoms in a way captured by the BPRS.This pattern resembles the "plateauing" of neuropsychiatric symptoms observed in advanced stages of Alzheimer's disease (Peters *et al.*, 2015), consistent with reduced behavioural responsiveness in severe stages. Individual variability was large, confirming strong heterogeneity in psychiatric trajectories seen in other cohorts (Rolland *et al.*, 2007). The random-effects structure (allowing each patient to have their own starting level and slope) showed considerable person-to-person variability, confirming the clinical impression that psychiatric trajectories in Alzheimer's disease are highly heterogeneous.

The patterns seen in the modelling were in line with the initial exploratory plots: both suggested a roughly linear increase in mean BPRS and strong within-patient correlation over time. The model does not exactly reproduce the raw variability at each visit, especially at later time points where the sample is smaller and more selected; however, this is expected and does not undermine the clinical interpretation, as the key trends and group differences are well captured.

Cognitive–functional decline (CDR-SB > 10)

For the binary outcome indicating severe cognitive/functional impairment (CDR-SB > 10), both patient-level and population-level perspectives were examined.The GLMM showed increasing odds of crossing into the severe CDR-SB category, consistent with established trajectories of cognitive decline (Morris *et al.*, 2001). Tau burden was associated with greater risk, echoing the strong link between tau pathology and clinical severity documented in tau-PET studies (Ossenkoppele *et al.*, 2016). This aligns with the clinical understanding that tau accumulation is closely linked to neurodegeneration and functional decline.

When we looked at population-averaged risk using a weighted GEE approach that corrects for dropout, the increase over time was more modest. This difference is expected: the patient-level model ("within-person" odds) and the population-level model (average risk in the cohort after adjusting for dropout) answer slightly different questions.The attenuation in weighted GEE reflects the fact that dropout removed higher-risk individuals from later follow-upa known phenomenon in dementia research (Schneider, Insel and Weiner, 2011). Together, they tell a coherent story: at the individual level, patients tend to progress toward more severe CDR-SB categories, and at the population level this progression is partly masked by selective dropout of the more impaired patients, which we attempted to correct for with weighting.

Sensitivity to MNAR (unobserved worsening)

Because we cannot directly observe what BPRS scores would have been after dropout, we performed a sensitivity analysis to mimic situations where patients who left the study might actually have had substantially worse symptoms than predicted under the MAR assumption. Even when we forced missing post-dropout BPRS values to be up to 10 points higher than the MAR-based predictions, the main conclusions did not change: BPRS still increased clearly over time, and the moderating role of baseline CDR-SB remained present and statistically convincing.

Clinically, this means that our key findings are robust to reasonably pessimistic assumptions about patients who left the study.This robustness aligns with methodological reports showing that LMM-based estimates of trajectories are often stable under reasonable MNAR deviations when dropout is clinically driven (Little, 1995) and (Wu and Carroll, 1988). The worsening of psychiatric symptoms over time and the influence of baseline cognitive/functional impairment are not fragile artefacts of our chosen statistical model; rather, they reflect stable patterns that persist under a range of plausible scenarios for the unobserved data.

Clinical implications

In summary, this analysis shows that psychiatric symptoms in Alzheimer's disease patients tend to worsen markedly over several year, with a high degree of individual variability. Baseline functional impairment, nursing-home residency, and other clinical factors identify patients who start at a higher symptom level and remain more burdened over time. Cognitive/functional severity at baseline shapes the trajectory of psychiatric symptoms, suggesting that symptom management and prognostic expectations need to be tailored to the patient's initial cognitive and functional status.

By using longitudinal models that explicitly account for dropout and by checking the robustness of conclusions under MNAR scenarios, we provide a more trustworthy picture than would be obtained from complete-case or naïve analyses. From a clinical standpoint, this supports the use of these modelling strategies to inform prognosis, to stratify patients in clinical trials, and to understand how symptom trajectories differ across clinically relevant subgroups in Alzheimer's disease.

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Appendix

Linear mixed-effects model for BPRS

```
## Linear mixed-effects model for BPRS (repeated for convenience)
library(lme4)
fm_adj <- bprs ~ time + sex +
    age + bmi + job + adl + wzc + cdrsb0 +
    time:cdrsb0 +
    1 + (time | patid) + (1 | trial)
lmer_reduced_mean_structure <- lmer(fm_adj,
    data = alzheimer_long_bprs,
    na.action = na.exclude
)
```

Generalized estimating equations for CDR-SB $>$ 10

```
library(data.table)
library(geepack)

# sort and create missingness indicators ---
setDT(alzheimer_long_cdrsb)
setorder(alzheimer_long_cdrsb, patid, time)

# Outcome indicator: 1 if observed, 0 if missing
alzheimer_long_cdrsb[, R := as.integer(!is.na(CDRSBbin))]
# Lagged history (within patient)
alzheimer_long_cdrsb[, `:=`(
  lag_R      = shift(R,        n = 1L, type = "lag"),
  lag_Y      = shift(CDRSBbin, n = 1L, type = "lag"),
  lag_abpet  = shift(abpet,    n = 1L, type = "lag"),
  lag_taupet = shift(taupet,   n = 1L, type = "lag")
), by = patid]

# At-risk rows for the discrete-time observation model:
# "among those observed at t-1, what is P(observed at t)?"
alzheimer_long_cdrsb[, at_risk := (time > 0 & lag_R == 1)]

#-Denominator model: P(R_it = 1 | observed history up to t-1) ---
den_fit <- glm(
  R ~ lag_Y + sex + age + bmi + adl + time + lag_abpet + lag_taupet,
  data   = alzheimer_long_cdrsb[at_risk == TRUE],
  family = binomial()
)

#Numerator model (stabilization): simpler P(R_it = 1 | baseline + time) ---
```

```r
num_fit <- glm(
  R ~ sex + age + bmi + adl + time,
  data   = alzheimer_long_cdrsb[at_risk == TRUE],
  family = binomial()
)

# Predicted conditional probabilities (set to 1 when not at-risk so cumprod is unchanged)
alzheimer_long_cdrsb[, `:=`(p_den = 1, p_num = 1)]
alzheimer_long_cdrsb[at_risk == TRUE, p_den := predict(den_fit, type = "response")]
alzheimer_long_cdrsb[at_risk == TRUE, p_num := predict(num_fit, type = "response")]

# Cumulative probabilities and stabilized weights ---
alzheimer_long_cdrsb[, `:=`(
  pi_den = cumprod(p_den),
  pi_num = cumprod(p_num)
), by = patid]

alzheimer_long_cdrsb[, w_stab := pi_num / pi_den]

# Keep weights only where outcome is observed (geeglm will drop missing outcomes anyway)
alzheimer_obs <- alzheimer_long_cdrsb[R == 1]

his_weights <- ggplot(alzheimer_obs, aes(x = w_stab)) +
  geom_histogram(bins = 30, color = "black", fill = "lightblue") +
  labs(
    title = "Stabilized Inverse-Probability-of-Observation Weights",
    x     = "Weight",
    y     = "Frequency"
  ) +
  theme_minimal()
his_weights


# Weighted GEE (WGEE):
wgee_model <- geeglm(
  CDRSBbin ~ time + sex + age + bmi +
    adl + abpet + taupet +
    time:abpet + time:taupet,
  id      = patid,
  data    = alzheimer_obs,
  weights = w_stab,
  family  = binomial(link = "logit"),
  corstr  = "ar1",
  std.err = "san.se"
)
```

Generalized linear mixed-effects model for CDR-SB > 10

```
alzheimer_long_cdrsb <- alzheimer_long_cdrsb |>
  dplyr::mutate(
    time_c   = time   - mean(time,   na.rm = TRUE),
    age_c    = age    - mean(age,    na.rm = TRUE),
    bmi_c    = bmi    - mean(bmi,    na.rm = TRUE),
    adl_c    = adl    - mean(adl,    na.rm = TRUE),
    abpet_c  = abpet  - mean(abpet,  na.rm = TRUE),
    taupet_c = taupet - mean(taupet, na.rm = TRUE)
  )

ctrl <- glmerControl(optimizer = "bobyqa", optCtrl = list(maxfun = 2e5))
glmm_center <- glmer(
  CDRSBbin ~ time_c + sex + age_c + bmi_c + adl_c +
    abpet_c + taupet_c + time_c:taupet_c +
    (time_c | patid),
  data   = alzheimer_long_cdrsb,
  family = binomial(link = "logit"),
  nAGQ   = 1,
  control = ctrl
)
```

PMM delta-shift sensitivity analysis for BPRS

```
# Prep
setDT(alzheimer_long_bprs)
setorder(alzheimer_long_bprs, patid, time)

# Dropout time D_i = last observed visit for BPRS (works even if missingness isn't perfectly monotone)
alzheimer_long_bprs[, D_i := max(time[!is.na(bprs)], na.rm = TRUE), by = patid]

# Indicator for "post-dropout" missingness: missing after the last observed BPRS
alzheimer_long_bprs[, post_dropout := is.na(bprs) & time > D_i]

# Function to build a delta-adjusted completed dataset and refit the same LMM
fit_pmm_delta <- function(delta, dat, mar_fit, fm) {
  dat_work <- copy(dat)

  # MAR-based predictions (conditional on estimated random effects)
  # re.form = NULL includes subject + trial random effects (BLUPs) for observed subjects/centres
  dat_work[, bprs_mar_hat := predict(mar_fit, newdata = dat_work, re.form = NULL, allow.new.levels = FALSE)]

  # Delta adjustment:
  # - For any missing BPRS, fill with MAR prediction
  # - If missingness is post-dropout (time > D_i), add delta
```

```r
  dat_work[, bprs_pmm := bprs]
  dat_work[is.na(bprs_pmm), bprs_pmm := bprs_mar_hat + delta * as.numeric(post_dropout)]

  # Refit the same mean/covariance structure on the delta-adjusted outcome
  fit_delta <- lmer(
    update(fm, bprs_pmm ~ .),
    data = dat_work
  )

  # Return fit + a tidy fixed-effect table
  beta  <- fixef(fit_delta)
  V     <- as.matrix(vcov(fit_delta))
  se    <- sqrt(diag(V))

  out <- data.table(
    delta = delta,
    term  = names(beta),
    estimate = as.numeric(beta),
    se = as.numeric(se)
  )
  out[, `:=`(
    lwr = estimate - 1.96 * se,
    upr = estimate + 1.96 * se
  )]

  list(fit = fit_delta, table = out)
}

# clinically plausible delta grid (BPRS points)
delta_grid <- c(0, 0.5, 1, 2, 3)

# Run the pattern-mixture delta sensitivity analysis
pm_list <- lapply(delta_grid, fit_pmm_delta,
                  dat = alzheimer_long_bprs,
                  mar_fit = lmer_reduced_mean_structure,
                  fm = fm_adj)

# Collect fixed effects across deltas
pm_fixed <- rbindlist(lapply(pm_list, `[[`, "table"))

#  (time trend + effect modification by baseline CDR-SB)
pm_key <- pm_fixed[term %in% c("time", "time:cdrsb0")]
```